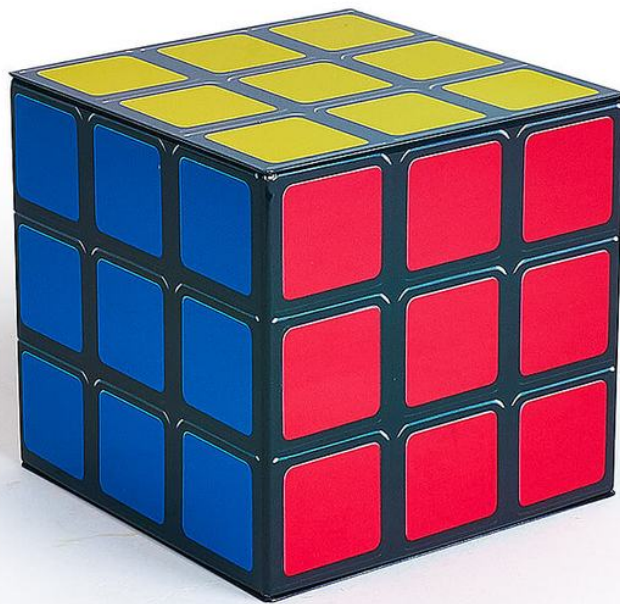


How to solve the Rubik's Cube Part 2: Advanced OLL and PLL algorithms



Notation



R



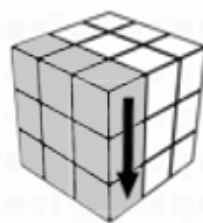
L



U



D



F



B



R'



L'



U'



D'



F'



B'



r



l



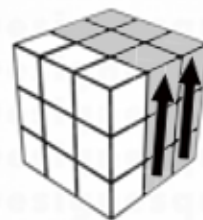
u



d



f



b



r'



l'



u'



d'



f'



b'



M



M'



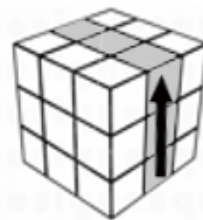
E



E'



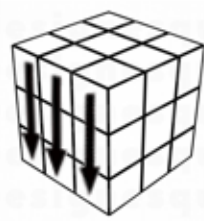
S



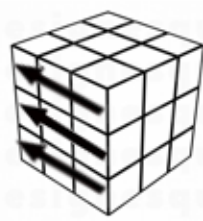
S'



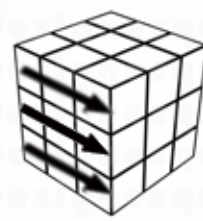
X



X'



Y



Y'



Z



Z'

Special Notation

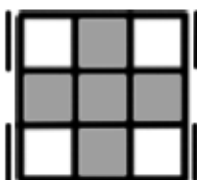
@	$R U R' U'$
!	$R' F R F'$
&	$R U R' U'$
<	$R U^2 R'$
@	$F @ F'$
@	$f @ f'$

@'	$L U' L' U'$
!'	$L F' L' F$
&'	$L' U' L U'$
<'	$L' U^2 L$
@ '	$F' @' F$
@ '	$f' @' f$

OLL

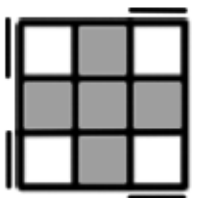
(Orient the Last Layer)

Cross



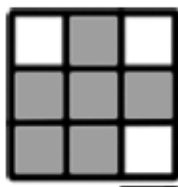
$\& (R U' R' U) <$

Tires



$||@|| |@|$

Flat Tires



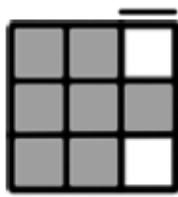
$\& <$

Right Tail Fish



$\&' <'$

Left Tail Fish



$(l U L' U') (R' U L U')$

Wheelbarrow



$(R^2 D) (R' U^2) (R D') (R' U^2) R'$

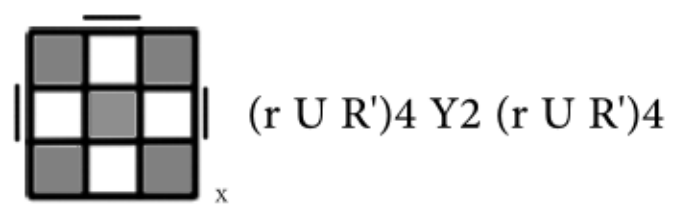
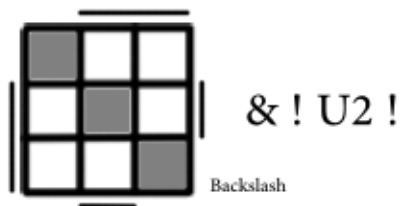
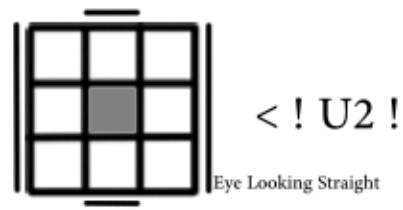
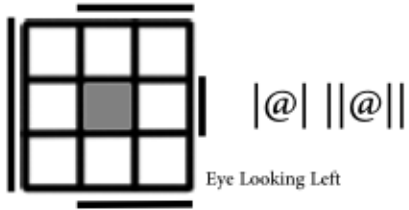
Headlights



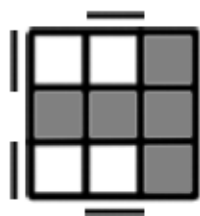
$X' (R' D' R U') (R' D' R U)$

Bow Tie

Dot

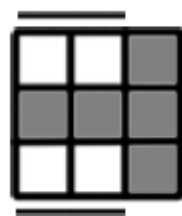


Line



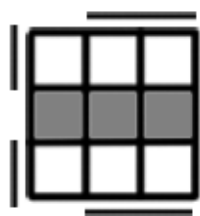
Nail 1

|@|



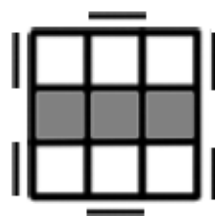
Nail 2

@ !



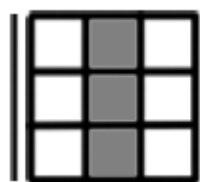
Blanket 1

||@||2



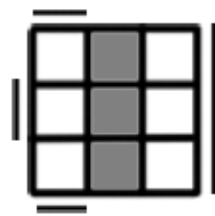
Blanket 2

(r U r') (U R U' R')² (r U' r')



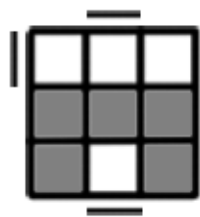
Banner 1

< (R' U' R U') (R' U²)
X (U R U')



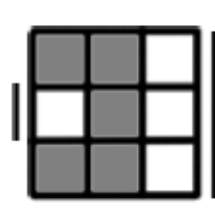
Banner 2

& (R d') (R U' R' F')



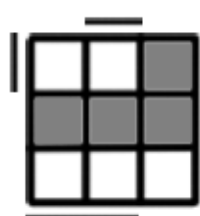
Crown

@ X' (U' R' D) (R E)



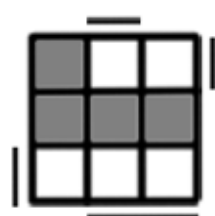
Punch

(R' U') ! (U R)



Right Couch

(r U r') @ (r U' r')



Left Couch

(l' U' l) @' (l' U l)



Right Gun

(r U') (L' B' L)
(U L' U') (f l)



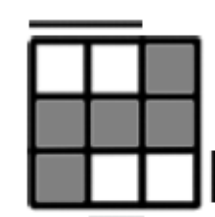
Right Gun

(l' U) (R B R')
(U' R U) (f' r')



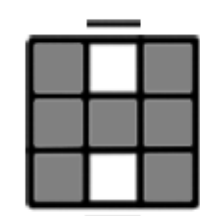
Right Lightning

R' |@| (U R)



Left Lightning

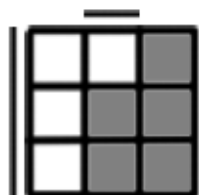
L |@|' (U' L')



H

@ M' (U R U') r'

Square



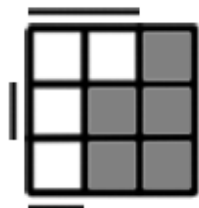
||@||

Right Chair 1



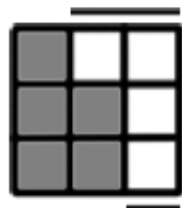
||@||'

Left Chair 1



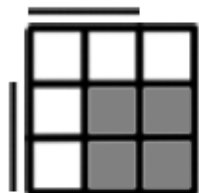
(R d L' d') (R' U R B R')

Right Chair 2



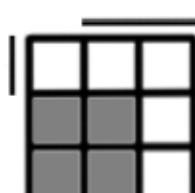
(L' d' R d) (L U' L' B' L)

Left Chair 2



r' U2 & r

Right Rock



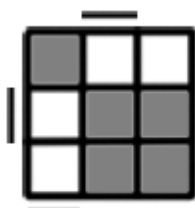
l U2 &' l'

Left Rock



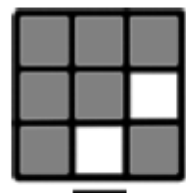
F (R U' R' U')
(R U R' F')

Soaring Kite



< ! <

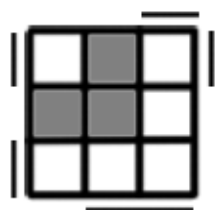
Falling Kite



(M U M') U2 (M U M')

Stingray

L



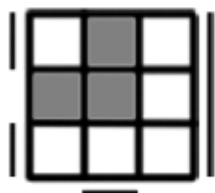
$|@|2$

Bottom-Bar Left-L



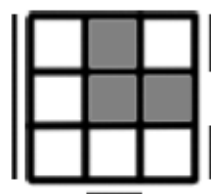
$|@|'2$

Bottom-Bar Right-L



$(R' F') !'2 (F l)$

Right-Bar Left-L



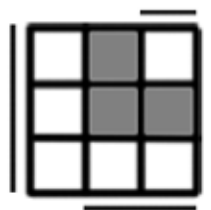
$(L F) !2 (F' r')$

Left-Bar Right-L



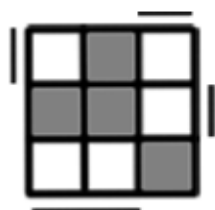
$X' (R' D) (R2 U')$
 $(R2 D') (R2 U) R'$

Two-Bar Left-L



$X' (L D') (L2 U)$
 $(L2 D) (L2 U') r$

Two-Bar Right-L



$@ (R' F R) @ F'$

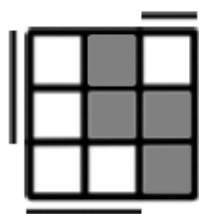
Right Shuttlecock



$@' (L F' L') @' F$

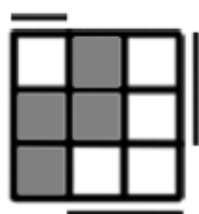
Left Shuttlecock

Z



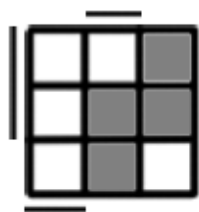
< U2 !

Right Attacking Cactus



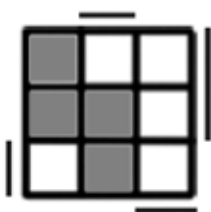
<' U2 !'

Left Attacking Cactus



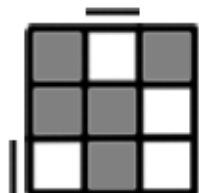
|@| U |@|

Right Defending Cactus



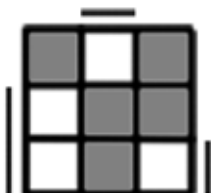
|@|' U' |@|'

Left Defending Cactus



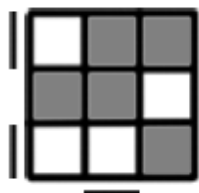
M U @ ! M'

Right Seal



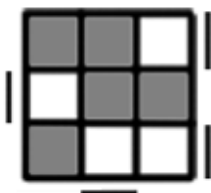
M U' @' !' M'

Left Seal



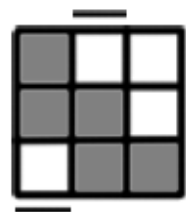
!2 @2

Right Gymnast



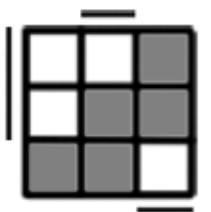
!'2 @'2

Left Gymnast



@ Y <' d @

Right Stairs



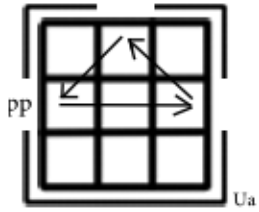
@' Y' < d' @'

Left Stairs

PLL

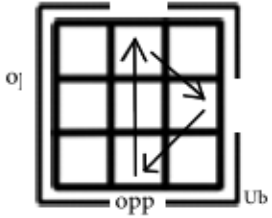
(Permute the Last Layer)

Edge Rotation



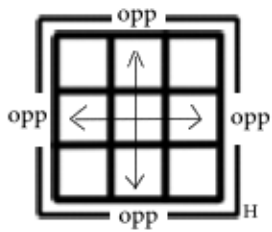
$(M2\ U)\ (M'\ U2)$

$(M\ U\ M2)$



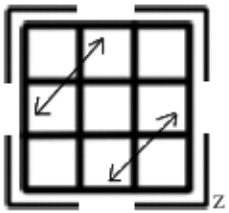
$R2\ (X'\ F)\ (U'\ D)$

$R2\ (U\ D')\ (X\ U)\ R2$



$(M2\ U)\ (M2\ U2)$

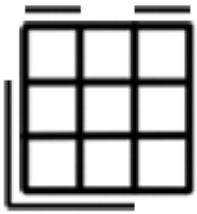
$(M2\ U)\ M2$



$(M2\ U)2\ (M\ U2)$

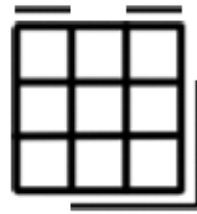
$(M2\ U2)\ M$

Headlights



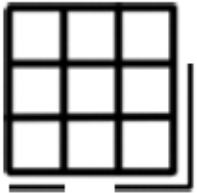
Aa

$X' (R' D) (R' U^2)$
 $(R D') (R' U^2) l^2$



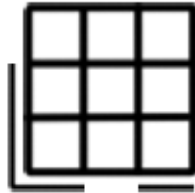
Ab

$X' (L D') (L U^2)$
 $(L' D) (L U^2) r^2$



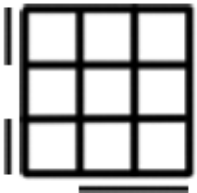
Ra

$(L U^2) (L' U^2)$
 $(L F') @' (r U) r^2$



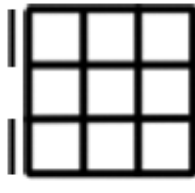
Rb

$(R' U^2) (R U^2)$
 $(R' F) @ (R' F') R^2$



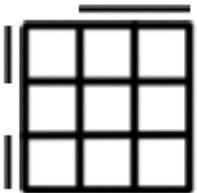
Ga

$R^2 (u R') (U R') (U' R)$
 $(u' R^2) Y' (R' U R)$



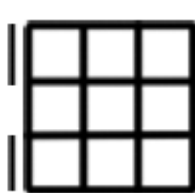
Gb

$(R' U' R) Y (R^2 u)$
 $(R' U) (R U') (R u') R^2$



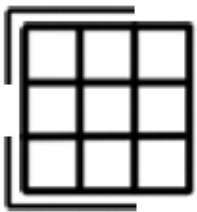
Gc

$R^2 (u' R) (U' R) (U R')$
 $(u R^2) Y (R U' R')$



Gd

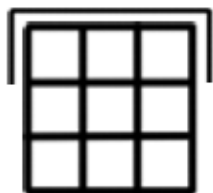
$(R U R') Y' (R^2 u')$
 $(R U') (R' U) (R' u) R^2$



T

$@ (R' F) (R^2 U')$
 $(R' U') (R U) (R' F')$

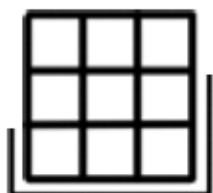
Long Bar



F

(R' U2) (R' d') (R' F')

(R2 U') (R' U) (R' F) (R U' F)



Ja

(R' U L') U2 (R U' R') U2 (R L)



Jb

(R U R' F') @

(R' F) R2 (U' R')

No Headlights, No Long Bar

